

ANDE PROJECT

Week 2 — Learning Journal

Adaptive Network Deception Engine

Everything we did, explained in simple terms.

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Week: Week 2 — Netty Server & First Connection

Goal: Make ANDE compile, run as a real server, and accept connections.

Result: SUCCESS — ANDE is listening on port 8080 and accepts client connections.

01. Starting Point

The project initially failed to compile because Main.java was in the wrong package, ConnectionHandler.java was missing, and there was no main() entry point.

02. Fixing the Project Structure

Moved Main.java into the com.ande.network package so the package declaration matched the folder structure.

03. Creating ConnectionHandler.java

Created ConnectionHandler.java to handle incoming client connections and resolve the 'cannot find symbol ConnectionHandler' compilation error.

04. Adding the Main Method

Added public static void main(String[] args) so Java could start the application and launch the Netty server.

05. Compiling with Maven

Ran 'mvn clean compile'. Learned that 'clean' removes old builds and 'compile' translates Java source into .class files. Build completed successfully.

06. Understanding Netty

Learned that Netty is a high-performance Java networking framework that manages sockets, threads, and client connections efficiently.

07. Running the Server

Started the application successfully. The terminal displayed 'ANDE listening on port 8080', confirming the server was waiting for connections.

08. Testing the Server

Connected to the server using a PowerShell TCP client because Telnet was unavailable. ConnectionHandler printed a successful client connection.

09. IPv4 vs IPv6

Learned that 127.0.0.1 (IPv4) and ::1 / 0:0:0:0:0:0:0:1 (IPv6) both represent localhost. Windows connected using IPv6 by default.

10. Achievement Summary

Resolved compilation issues, created ConnectionHandler, added a main() method, compiled successfully with Maven, started a Netty server, opened port 8080, and accepted the first client connection.

Week 3 Preview

Next week ANDE will detect protocols (SSH, HTTP, FTP, Telnet), log attacker activity, and begin emulating network services.